

- 1 Fig. 15 shows a uniform shelf AB of weight WN .
The shelf is 180 cm long and rests on supports at points C and D. Point C is 30 cm from A and point D is 60 cm from B.

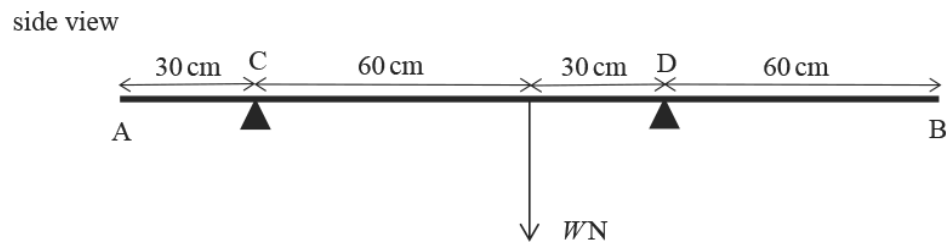


Fig. 15

Determine the range of positions a point load of $3W$ could be placed on the shelf without the shelf tipping. [6]

- 2 A non-uniform rod 0.8 m long rests horizontally on smooth pegs A and B at each end of the rod. The contact forces at A and B are 10 N and 15 N respectively, as shown in Fig. 2.

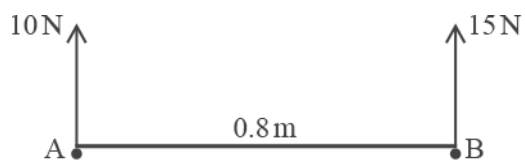


Fig. 2

Calculate the distance of the centre of mass of the rod from A.

[3]

- 3 A rod of length 2 m hangs vertically in equilibrium. Parallel horizontal forces of 30 N and 50 N are applied to the top and bottom and the rod is held in place by a horizontal force F N applied x m below the top of the rod as shown in Fig. 7.

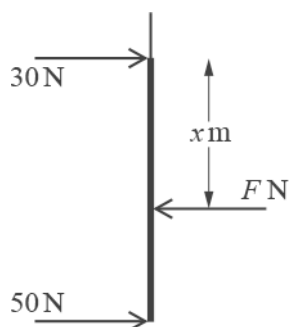


Fig. 7

- (a) Find the value of F .

[1]

- (b) Find the value of x .

[2]

4 Fig. 7 shows a rectangular lamina ABCD with sides AB of length 60 cm and AD of length 50 cm. The lamina is lying flat on a smooth horizontal surface, and is acted on by the following five horizontal forces.

- 45 N at A in the direction BA
- 40 N at D in the direction AD
- 27 N at C in the direction BC
- Y N at A in the direction DA
- X N at E in a direction perpendicular to the edge BC

The lamina is in equilibrium. The distance BE is d cm.

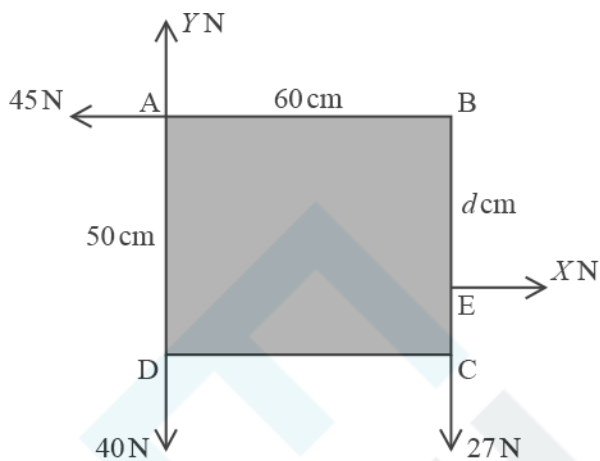


Fig. 7

Calculate the values of each of the following.

- X
- Y
- d

[4]

- 5 Fig. 3 shows a rod AB which is 0.9 m long and hangs vertically from a smooth hinge at A. The rod can rotate about A in a vertical plane. Forces of 100 N and 60 N act at right angles to AB in this plane. Their points of application are 0.3 m and 0.75 m respectively below A.

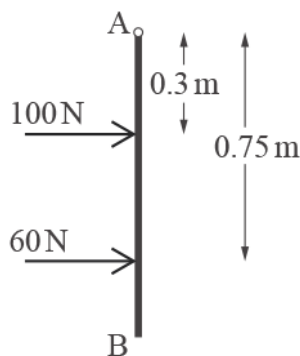


Fig. 3

- (a) Find the combined moment of these forces about A.

[2]

The rod is held in equilibrium by a force of F N which is also at right angles to the rod in the same vertical plane.

- (b) Find the least possible value of F .

[2]

- (c) Explain how the rod can be in equilibrium when the resultant of these three forces is not zero.

[2]

- 6 Fig. 4 shows a uniform beam of mass 4 kg and length 2.4 m resting on two supports P and Q. P is at one end of the beam and Q is 0.3 m from the other end.

Determine whether a person of mass 50 kg can tip the beam by standing on it.

[3]

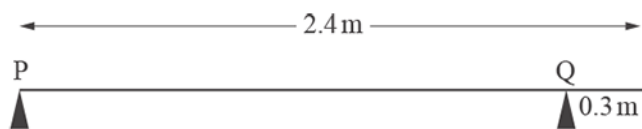


Fig. 4

- 7(a) A uniform ruler AB has mass 28 g and length 30 cm. As shown in Fig. 6, the ruler is placed on a horizontal table so that it overhangs a point C at the edge of the table by 25 cm.

A downward force of F N is applied at A. This force just holds the ruler in equilibrium so that the contact force between the table and the ruler acts through C.



Fig. 6

Complete the force diagram below, labelling the forces and all relevant distances.

[2]



- (b) Calculate the value of F .

[2]

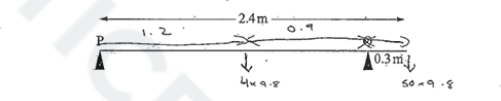
END OF QUESTION PAPER

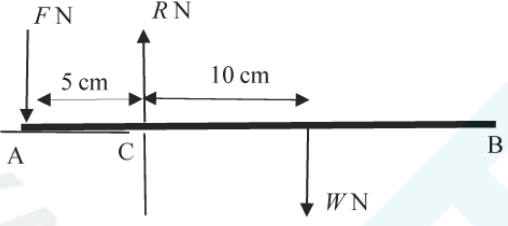
Question			Answer/Indicative content	Marks	Guidance
1			Let the reactions of the supports on the shelf be U N at C and V at D Neither U nor V can be negative if the shelf does not tip Any position between C and D must give $U > 0$ and $V > 0$ Consider putting the load between A and C, x cm from C cw moments about C If $V \geq 0$ then $W \times 60 \geq 3W \times x$ so $x \leq 20$ Consider putting the load between D and B, y cm from D anti-cw moments about D If $U \geq 0$ then $W \times 30 \geq 3W \times y$ so $y \leq 10$ The load must be placed not closer than 10 cm to A and 50 cm to B oe	M1(AO2.2a) B1(AO2.2a) M1(AO3.1b) A1(AO1.1) B1(AO3.1b) A1(AO3.2a) [6]	May be implied Need not show but must be stated moments about C allow < allow < Must be clear statement and include CD
			Total	6	
2			Vertical equilibrium: weight = $10 + 15 = 25$ N Moments about A: $25x = 15 \times 0.8$ $x = 0.48$ m	B1(AO1.1a) M1(AO3.1b) A1(AO1.1b) [3]	soi or take moments about other points
			Total	3	
3		a	$F = 30 + 50 = 80$ N	B1 (AO1.1a) [1]	Cao Examiner's Comments This was a routine mark for which almost all candidates were credited, the only mistake seen was to find the difference between 50 N and 30 N,

Question			Answer/Indicative content	Marks	Guidance	
		b	Taking moments about the top of the rod $Fx = 50 \times 2$ $x = 1.25 \text{ m}$	M1 (AO 3.3) A1 (AO 1.1b) [2]	Or any other suitable point Cao <u>Examiner's Comments</u> Most candidates successfully took moments about the top of the rod to easily obtain the correct answer. Candidates that attempted to take moments about the centre of the rod often encountered problems with their positioning of F with respect to the centre.	All necessary terms must be present. Each term must be a product of a force and a length.
			Total	3		
4			Resolve $\rightarrow$: $X - 45 = 0 \Rightarrow X = 45$ Resolve $\uparrow$: $Y - 27 - 40 = 0 \Rightarrow Y = 67$ Moments about A: $Xd - 27 \times 60 = 0$ $d = \frac{27 \times 60}{45} = 36$	B1 (AO 1.1a) B1 (AO 1.1a) M1 (AO 3.4) A1 (AO 1.1) [4]	oe; any moments equation must be correct, apart from sign errors FT their value for X	Answer for d must correspond to cm units; do not accept 0.36 or 0.36m.
			Total	4		

Question			Answer/Indicative content	Marks	Guidance	
5		a	Taking moments about A: $0.3 \times 100 + 0.75 \times 60$ $= 75 \text{ Nm anticlockwise}$	M1 (AO 1.1b) A1 (AO 1.1b) [2]	Allow for one moment only, but if two terms present, both must be moments cao; direction needed	Do not allow for the total of a force and a moment
		b	Minimum force when applied at B: $0.9 F = 75$ $F = 83.3$	M1 (AO 3.1b) A1 (AO 1.1b) [2]	Forming equilibrium equation with distance 0.9 FT their moment in (a)	
		c	The hinge provides an additional [horizontal] force at A But this will not have a moment about A	B1 (AO 3.2a) B1 (AO 3.2a) [2]	Must mention either A or the hinge Must mention moment about A	Do not allow either mark for an argument based on the weight
			Total	6		

Question	Answer/Indicative content	Marks	Guidance
6	Let R_P and R_Q be the normal reactions upwards Assume the person stands on the end beyond Q and that the beam is in equilibrium Taking moments about Q: $50g \times 0.3 + R_P \times 2.1 = 4g \times 0.9$ $R_P = \frac{3.6g - 15g}{2.1} [= -53.2] < 0$ So the beam will tip. Alternative solution If the person stands x m beyond Q so that the beam is on the point of tipping $R_P = 0$ Taking moments about Q $4g \times 0.9 = 50gx \text{ giving } x = 0.072$ $0.072 < 0.3$ so this is possible while standing on the beam Second alternative solution Largest clockwise moment of the weight of the person about Q is $50g \times 0.3 + 15g$ [= 147] Anticlockwise moment of the weight of the beam is $4g \times 0.9 = 3.6g$ [= 35.28] The moment of the person's weight is larger so the beam will tip	M1 (AO 3.1b) A1 (AO 1.1b) A1 (AO 3.2a) [3] M1 A1 A1 M1 A1 A1	Finding moment of a force about any point Correct equation from moments about Q Conclusion must be clear from correct working (R_P need not be evaluated but must be clear that it is negative) Finding moment of a force about any point Correct equation from moments about Q Conclusion must be clear from correct working and reference to 0.3 m Finding moment of a force about any point Both correct moments about Q required Conclusion must be clear from a comparison of moments of two forces If the weight of the person shown between P and Q allow M1 maximum. Could also be obtained from moments about P and resolving to evaluate R_P SC 2(omitting g) $0.9 \times 4 < 0.3 \times 50$ $3.6 < 15$ So the beam will tip. <u>Examiner's Comments</u>

Question	Answer/Indicative content	Marks	Guidance
			This more open-ended question was not well answered as many candidates did not write a convincing argument using the results of their calculations. Most realised moments were required and the first method mark was given for any use of moments. Some candidates omitted g in their calculations. This question was most easily answered by taking moments about Q, the point about which the beam would turn if the person stood in a suitable place. A correct solution can also be constructed from moments about P, to give a value for the reaction at Q that was larger than the combined weights of the beam and the person, then arguing that the beam would need to be held down at P to maintain equilibrium. Candidates often did not explain where their equations or terms came from or what they signified. Full marks were only given where sufficient explanation was also seen.  Misconception Some candidates found the value of the reaction at either P or Q for the beam alone without realising that the reactions will be different when the person stands on the beam. Exemplar 2  $400 \times 1.2 \times 0.9 = 35.28 \text{ moments}$ $500 \times 0.3 \times 0.3 = 47 \text{ moments}$ the person can tip the beam

Question			Answer/Indicative content	Marks	Guidance
					This exemplar shows the importance of making explicit the link between the mathematical working and the conclusions to be drawn. Here they have written some correct arithmetic but not explicitly stated that their conclusion is based upon comparing these values (the anticlockwise and clockwise moments about Q) to show that the person's weight will tip the beam.
			Total	3	
7	a			B1(AO1.1a) B1(AO1.1) [2]	F and their weight even if not labelled in roughly correct position Distances labelled. Allow for 5cm and any measurements that show that the weight is acting in the centre of the ruler. R is already drawn in Printed Answer Booklet
	b		Take moments about C $10 \times 0.028g = 5F$ $F = 0.549 \text{ N}$	M1(AO1.1a) A1(AO1.1) [2]	Moments about any point with all relevant forces seen. Allow their weight and one incorrect distance [Value of R is 0.8232 N] allow $\frac{7}{125}g$ oe
			Total	4	